[T]-Theory: Economics

P15 · Capstone · For: The Economist

The Nash Attractor Resolvent — markets as Hopfield networks settling into equilibria.

Economy as field theory:

- Market = Hopfield network in 8D agent strategy space
- Nash equilibrium = fixed point of step(W_{market})
- Economic shock = perturbation of the collective field
- Recession = field trapped in a suboptimal attractor well

Key identifications:

- Price signal = field gradient ∇H
- Interest rate = field temperature T_{field}
- Central bank = volitional injection $J_{\text{CB}}(t)$
- Inflation = field running away from equilibrium

I · The Nash Attractor Resolvent

The Nash equilibrium as a fixed point of the economic propagator:

$$G_{\text{Nash}}(\lambda) = (H_{\text{market}} - \lambda I)^{-1}$$

A Nash equilibrium corresponds to a pole of G_{Nash} at λ^* where $\lambda^* = \min H_{\text{market}}(\Psi)$.

The **resolvent** determines which equilibrium the market settles into, given initial conditions — exactly the Clinical Operator Propagator (P10) in an economic substrate. Same mathematics; different actors.

II · Market Dynamics as Field Evolution

$$\dot{\Psi}_{\text{market}} = -\nabla H_{\text{market}}(\Psi) + J_{\text{policy}}(t) + \eta_t$$

Where:

- $H_{\text{market}}(\Psi) = -\frac{1}{2}\Psi^T W_{\text{market}}\,\Psi$ (Hopfield form)
- $J_{\text{policy}}(t)$ = monetary/fiscal intervention
- η_t = market noise (temperature $T_{\text{field}} \propto$ volatility)

This is Arrow–Debreu general equilibrium theory re-expressed as gradient flow.

III · Economic Shocks as Field Perturbations

- Supply shock = W_{market} coupling matrix changes suddenly
- Demand shock = source term $J(t)$ spike
- Financial crisis = field escaping a basin, cascading through the network
- Recovery = field finding a new Nash attractor (often suboptimal)

Secular stagnation = field in a low-energy basin with high escape barrier (low interest rate $\neq$ low temperature in this model).

IV · Welfare as Negative Energy

Aggregate welfare:

$$W = -H_{\text{market}}(\Psi^*) = \frac{1}{2}\Psi^{*T} W_{\text{market}}\,\Psi^*$$

Maximising welfare = finding the global minimum of $-H$. Policy = reshaping W_{market} to deepen the welfare-optimal well.

G-ID: *The Nash Attractor Resolvent* — $(H-\lambda)^{-1}$ determining market equilibrium. ORCID: 0009-0007-2194-0850 · · Zurich 2026